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Classic Logic Gates Overview

Video: Classic Logic Gates Overview
6 min

Graded Assignment: Assessment 2
Submitted

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Next item

Assessment 2

Review Learning Objectives

1. Which devices are used in classical computers for storing bits? Select all that apply. **1 / 1 point**

- [x] Solid-state drives

  Correct
  Correct! Solid-state drives are widely used for data storage in classical computers.

- [ ] Quantum dots
- [ ] Josephson junctions
- [x] RAM

  Correct
  Correct! RAM is another common storage device in classical computers.

- [x] Hard drives

  Correct
  Correct! Hard drives are commonly used to store bits in classical computers.

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Assignment details

Due
Dec 13, 11:59 PM IST

Attempts
Unlimited

Submitted
Dec 10, 12:38 AM IST

Retry

2. What is a Josephson junction? **1 / 1 point**

- ( ) A semiconductor device used in classical computers
- ( ) A type of quantum algorithm
- (•) A superconducting device that can detect extremely low magnetic fields
- ( ) A data storage unit in quantum computers

Correct
Correct! Josephson junctions are indeed superconducting devices used for detecting extremely low magnetic fields.

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3. What is one of the primary challenges in maintaining quantum coherence in quantum circuits? **1 / 1 point**

- ( ) Limited storage capacity
- ( ) High computational power
- ( ) Slow processing speeds
- (•) Environmental noise

Correct
Correct! Environmental noise can cause quantum decoherence, thereby disrupting the quantum state.

4. Given the quantum state vector **1 / 1 point**

$$\psi = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

, what is the probability of measuring the state in the basis vector $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$?

- ( ) 0
- ( ) 1
- ( ) 0.25
- (•) 0.5

Correct
Great job! The probability is given by the square of the magnitude of the projection of ψ onto $\mathbf{e}_1$.

5. Which of the following logic gates outputs true only if both of its inputs are true? **1 / 1 point**

- ( ) OR gate
- ( ) NOT gate
- (•) AND gate
- ( ) NAND gate

Correct
Correct! The AND gate outputs true only if both of its inputs are true.